

PROBLEM-1

To estimate the true error rate of hypothesis $h_{\frac{1}{2}}$ on the underlying population, we use the error observed on the test set.

Given:

- Number of test instances:

$$n = 45$$

- Observed error rate:

$$\hat{e} = 6.67\% = 0.0667$$

formula for 95% confidence interval

The confidence interval for the true error is given by:

$$\hat{e} \pm z \sqrt{\frac{\hat{e}(1-\hat{e})}{n}}$$

for 95% confidence interval, $z = 1.96$

- Compute standard error

$$= \sqrt{\frac{0.0667(1-0.0667)}{45}}$$

$$= \sqrt{\frac{0.0667 \times 0.9333}{45}}$$

$$= \sqrt{\frac{0.0622}{45}}$$

$$= \sqrt{0.00138} = 0.0372$$



→ Compute the margin error

$$1.96 \times 0.0372 = 0.0729$$

→ find confidence interval

lower limit

$$0.0667 - 0.0729 = -0.0062$$

upper limit

$$0.0667 + 0.0729 = 0.1396$$

Error rate cannot be -ve, so the lower bound is taken as 0.

Thus, the 95% confidence interval for the true error is:

$$(0, 0.1396)$$